An 18 b 12.5 MS/s ADC With 93 dB SNR

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***Abstract*—This paper presents a precision 18-bit 12.5 MS/s ADC that was designed primarily for digital X-ray imaging systems. This ADC was intended to have a faster output data rate than the precision successive approximation ADCs normally chosen for these systems but with similar DC accuracy and dynamic range. The chosen architecture consists of a pipeline of two multi-bit successive approximation converters. The first successive approximation ADC generates an initial coarse conversion result. The DACs within this converter are then used to generate a residue which is amplified by a residue amplifier before being converted by a second successive approximation ADC. Four comparators within each ADC allow 2 bits to be determined each bit trial. Capacitor mismatch errors are digitally corrected with error coefficients stored in non-volatile memory. Dither is used to reduce the effect of errors in the flash ADC within the second ADC. The ADC was implemented on 0.25μm CMOS process with PIP capacitors and achieves a SNR of 93 dB with a 50 kHz input tone. INL and DNL are within ± 2.2 LSB and ± 0.3 LSB respectively. Power consumption is 105 mW, excluding LVDS interface power.**

***Index Terms*—ADC, calibration, dither, flash, pipeline, successive approximation.**

I. Introduction

DIGITAL X-ray image capture is increasingly being chosen for medical diagnostics over traditional X-ray film as it provides the potential benefits of reduced patient exposure, digital enhancement of the image, faster patient throughput and lower image archiving costs. In digital X-ray systems each pixel in a photodiode array is processed by a charge integrator, the outputs of which are multiplexed to a number of ADCs. These ADCs need to have both a high dynamic range, typically better than 90 dB and excellent linearity. The dynamic range requirement is due in part to the phenomenon that the number of X-ray photons reaching the sensor decreases exponentially with the distance that the X-rays pass through the patient: just 20 mm of soft body tissue can be enough to reduce the number of transmitted X-rays by half (6 dB). The need for high linearity is due to the way that different areas of the photo-diode array are processed by different ADCs. To avoid visible artifacts at the boundaries between these areas, all the ADCs need to have matching linearity. This is best ensured by placing a tight linearity requirement on each ADC, with linearity better than 15 ppm being generally specified.

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Pipeline converters are generally optimized for high SFDR. High SFDR can be more readily achieved with a relatively small input signal, typically no more than 2 V pk-pk differential, where the input switches and driving circuits are operating in a more linear region. Few pipeline converters achieve better than 80 dB SNR or DR for this reason. Capacitor matching errors along with residue amplifier (RA) gain, settling and linearity errors can be significant sources of ADC linearity error. Digital calibration of capacitor errors and dither have been used in [1] to achieve linearity of 22.9 ppm which is close to the linearity requirement for this application. Delta-sigma ADCs do exhibit sufficiently high dynamic range and excellent linearity, however, the digital filters needed to remove their quantization noise have settling times that are too long to give acceptable channel switching rates. The required dynamic range and linearity requirements are met by a number of commercially available 16-bit successive approximation register (SAR) ADCs at reasonable conversion rates.

To increase SAR ADC conversion rates, efforts have been made to decrease the bit trial time, of which the DAC settling time is usually the largest component. Fixed radix < 2 DAC bit weights [2] or additional redundant bits [3] within the DAC have allowed errors made by strobing the comparator latch before the DAC has fully settled to be corrected. This built in tolerance to incomplete DAC settling has allowed the time allocated to DAC settling to be reduced.

Further improvements in SAR conversion rate have been achieved by determining 2 or more bits per bit trial by either using multiple DACs and comparators [4] or a single DAC and a low resolution flash ADC [5]. The 6-bit ADC described in [4] uses 3 DACs and 3 comparators to generate and test the 3 levels required to determine 2 bits each bit trial. However, different comparator offset errors contribute directly to the non-linearity of the SAR ADC and as a result this approach has been limited to low to medium resolution ADCs. In [6] an RA is used to attenuate the linearity errors of a 4-bit flash ADC. A single DAC is used to generate a residue that is amplified by a RA before being converted by this flash ADC. The linearity errors of the flash ADC are attenuated by the RA gain (when referred to the input of the ADC) and this then allows a higher resolution to be achieved, in this case 14-bits. One drawback of this approach is that the RA is within the successive approximation loop and must be allowed to settle to sufficient accuracy each bit trial.

This paper describes an 18-bit ADC that uses a RA to amplify a single residue generated by DACs within a primary SAR ADC. This amplified residue is then sampled and converted by a secondary SAR ADC. The results from the 2 ADCs are combined to produce a final result. The primary ADC uses 4 comparators and the secondary ADC that converts the RA output uses a 4 level flash ADC to allow 2 bits to be determined during each (but the last) bit trial. The settling delay associated with the RA

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occurs just once per conversion rather than during each bit trial as in [6]. The gain of the RA is set high enough to ensure that the linearity errors of the flash within the secondary ADC, when referred to the ADC input, are typically less than 1 LSB at the 18-bit level. This architecture is similar to the pipeline of conventional successive approximation converters described in [7] except in this work 2 bits are determined during all but the last bit trial.

Section II starts by describing why SAR ADCs exhibit high accuracy and low power but are relatively slow. Multi-bit bit trials, redundancy and pipelining techniques are then discussed that can be used to increase the speed of SAR ADCs. Section III describes the architecture of the ADC and Section IV describes the implementation of the individual circuit blocks. Section V presents the measurement results and Section VI concludes the paper.

II. Background Information

A. Successive Approximation Converters

Traditional SAR ADCs are known to be linear and power efficient but, due to the serial way in which each bit is determined, are not architected for high throughput rate. Their linearity arises from the way that the comparator is used to simply determine the sign of the DAC output relative to the input. As long as the comparator does not exhibit hysteresis or any other memory effects and the DAC is given sufficient time to fully settle, the DC linearity of the converter is determined solely by the linearity of the DAC. This contrasts with pipeline converters where the linearity and gain error of the RA, particularly in the first MDAC stage, also contribute to the non-linearity of the converter.

The high power efficiency that has been achieved with SAR ADCs is mainly due to the lack of any feedback amplifiers within the architecture. In low resolution converters the comparator regenerative latch typically takes no quiescent power. In medium and high precision converters the latch is preceded by one or more preamps that reduce the input referred offset and noise of the latch [8]. However these preamps do not operate in feedback, as the precise gain of the preamp does not need to be controlled and as a result they can be implemented in a very simple and power efficient manner.

B. Multi-Bit Successive Approximation ADCs

Unlike pipelined converters, traditional SAR ADCs only determine one bit per quantizing time step. By using multiple comparators, more than one bit can be determined at each trial step. Fig. 1 shows an example of a conversion cycle for a 4-bit SAR converter where 2 bits are determined during each of the 2 bit trials. Three DACs and three comparators can be used to generate and test the 3 different test levels. Fig. 2 shows the code transitions for the same converter with an offset added to the comparator that tests the centre level of each search space during both bit trials. This offset has the effect of producing uneven code widths in the ADC transfer function. Unlike a traditional SAR ADC, where comparator offset only gives an offset

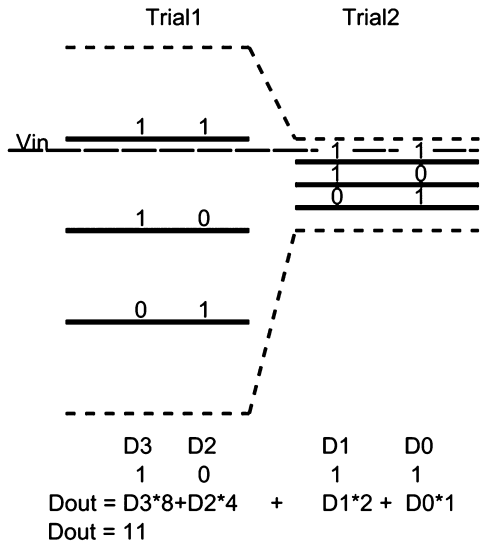


Fig. 1. Conversion cycle for 4-b ADC with 2 bits determined each bit trial.

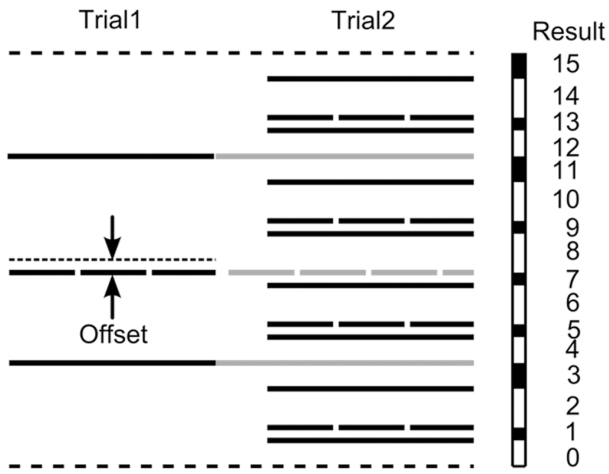


Fig. 2. Code transitions for 4-b ADC with offset applied to centre comparator.

in the overall transfer function, different comparator offsets in a multi-bit SAR ADC now directly degrade the linearity of the converter.

C. Redundancy

To allow correct conversion with incomplete DAC settling, it is known to use a DAC with a $\mathrm{fixed\ radix} < 2$ or with non-constant radix, by including extra redundant bits. Fig. 3 shows an example conversion cycle of the 4-bit converter of Fig. 1 adapted to provide extra range during the second bit trial Trial2, by adding a redundant level D0r. The input is at a voltage equivalent to 4.25 LSB. To allow correction of settling errors of either sign, all the test thresholds during the last bit trial have been shifted down by half a LSB. The DAC that generates the 0100 test level during Trial1 is still above $\mathrm{V_{in}}$ when the comparator is strobed and so this level is incorrectly rejected. During Trial2, the extra range provided by the redundant bit D0r allows this error to be recovered: The result of 3.5 being correct to within

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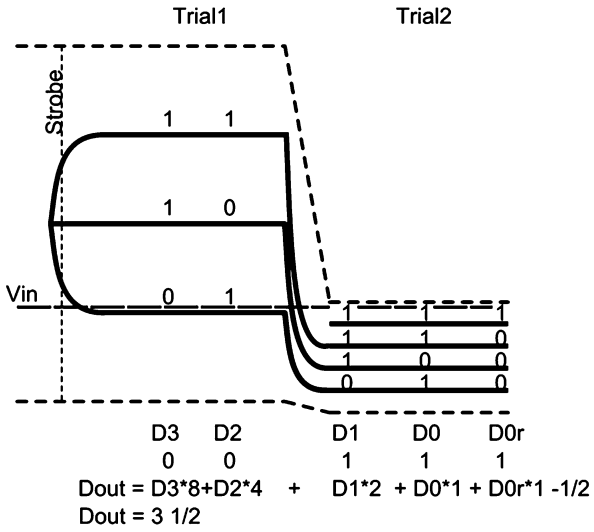


Fig. 3. Conversion cycle with redundant test level added during Trial2.

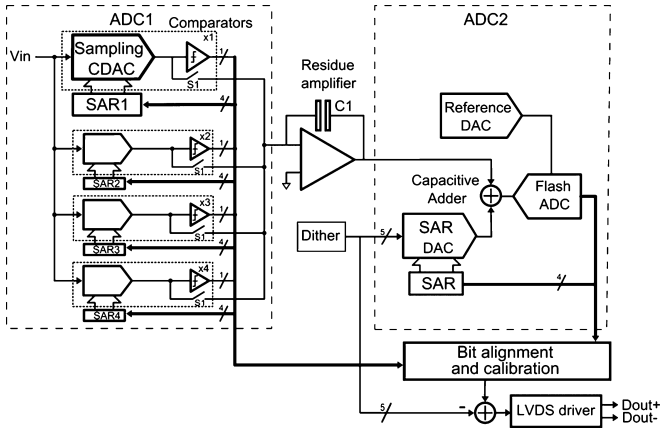


Fig. 4. Block diagram of the 18-bit pipelined SAR converter (actual implementation is differential).

the resolution of the converter. Without redundancy, just D1 and D0 would be kept, giving a result of 3 which is incorrect.

D. Pipeline of Successive Approximation ADCs

Multi-bit bit trials can be used throughout a conversion as long as the offsets in the comparators (and between the different DACs generating the different test levels) are small relative to the LSB size of the converter. For this work, the converter has an 8.2 V peak to peak input; these comparators would need to be designed to have offsets less than 31 μV.

An alternative to making the comparator offsets smaller is to use an amplifier to increase the size of the signal presented to the comparators during a second part of the conversion. This can be achieved, without saturating the outputs of this amplifier, by presenting the amplifier with a small residue, equal to an accurate difference between the input and a coarse estimate of the input, determined during the first part of the conversion. The LSB size of the ADC, referred to the output of this RA, is now increased by the RA gain. The comparators in a multi-bit SAR ADC, used to convert the output of the RA, can now have offsets that are larger by a factor equal to the gain of the RA. With sufficient gain in the RA, multi-bit bit trials can then be used in both the first converter that generates the coarse estimate of the input required to generate the residue, and also in the second converter that converters the output of the RA.

III. ADC Architecture

The architecture of this 18-bit pipelined multi-bit SAR ADC is shown in Fig. 4. ADC1 generates a coarse 10-bit estimate of the input over 5 bit trials and generates a small residue using the same sampling DACs used to perform this initial conversion. The closed loop RA amplifies this residue by a factor of 42.66. ADC2 samples and converts the output of the RA, generating a 9-bit result (with one bit of overlap across the RA output) over 5 bit trials. To correct for capacitor bit weight errors, the outputs from ADC1 and ADC2 are digitally corrected during the conversion, using error coefficients measured at final test and stored in non-volatile memory. The corrected outputs from both ADCs are then merged and clipped to provide a final result. Dither is used within ADC2 to reduce the magnitude of DNL errors caused by offsets in the ADC2 comparators.

Once ADC2 has sampled the RA output, the ADC1 sampling DACs are switched back to the ADC input to start acquiring the next input sample. The time constant of these DACs is just 1.7 ns which is short enough to allow the input to be sufficiently acquired by the end of the ADC2 conversion. As a result, immediately after ADC2 has completed the current conversion, ADC1 sample switches can be opened and a new ADC1 conversion started. This contrasts with a traditional SAR ADC where a signal acquire phase is required between the end of one conversion and the start of the next.

IV. Circuit Details

A. ADC1

A block diagram of ADC1 is shown in Fig. 5. The input voltage is sampled on to 4 identical sampling DACs (each with a capacitance of 6.25 pF) each with its own comparator. To minimize distortion at higher input frequencies, the gates of the left hand side sampling switches are bootstrapped to the input [9], [10]. The 4 SARs differ and are designed to allow each of the 4 DACs to be set to 4 different, but equally spaced levels during each ADC1 bit trial. While only 3 test levels are required to determine 2 bits, the 4th level provides redundancy, allowing some recovery from errors made in previous bit trials, such as errors due to incomplete DAC settling. The result from each bit trial, as indicated by the comparator outputs, is used to set the 2 bits (plus the redundant bit) under test to the same state in all 4 ADCs.

Fig. 6 shows the DAC switch settings for the first bit trial. The different switch setting for the top 2 bits C16 and C17 in the lower 3 DACs allow the input to be tested against the (1/4), (1/2), and (3/4) full-scale levels as shown in Fig. 8. The top DAC is not used for the first bit trial. If, as in this example, the input lies between (1/2) and (3/4) full scale the C17 MSB capacitor (driven terminal) will be switched to V_{refp} and C16 will be switched to V_{refn} for the second bit trial, as shown in Fig. 7. Capacitors C15, C14 and C14r will also now be set to 4 different states within the 4 different DACs and along with

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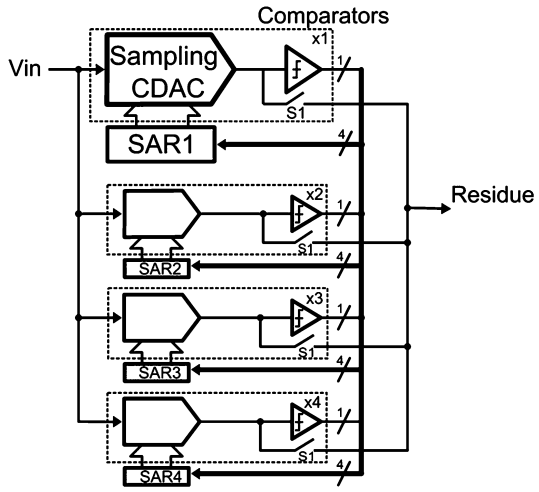


Fig. 5. Block diagram of ADC1 (actual implementation is differential).

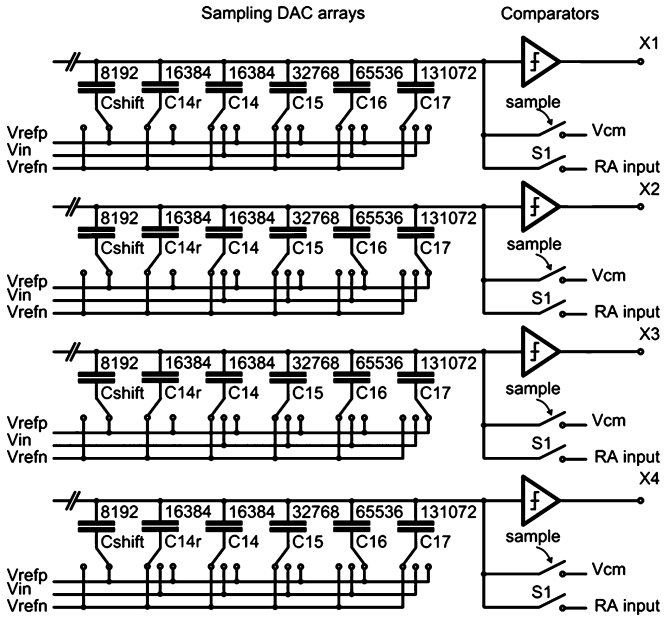


Fig. 6. DAC settings for ADC1 first bit trial.

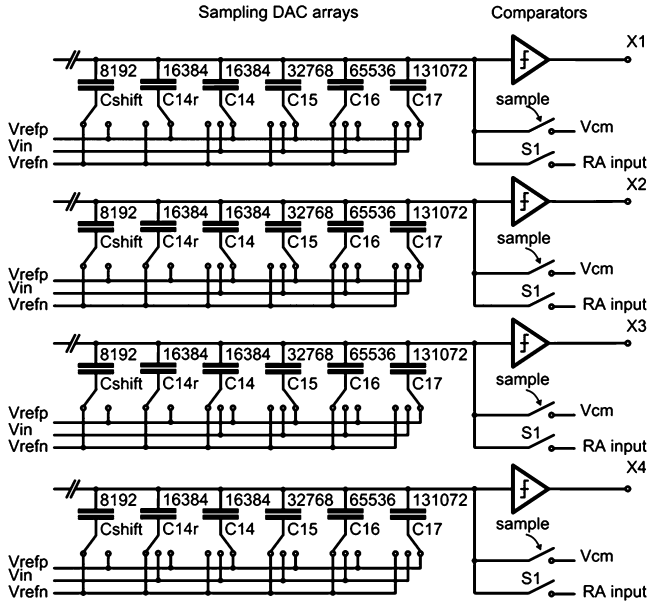


Fig. 7. Example DAC settings for ADC1 second bit trial.

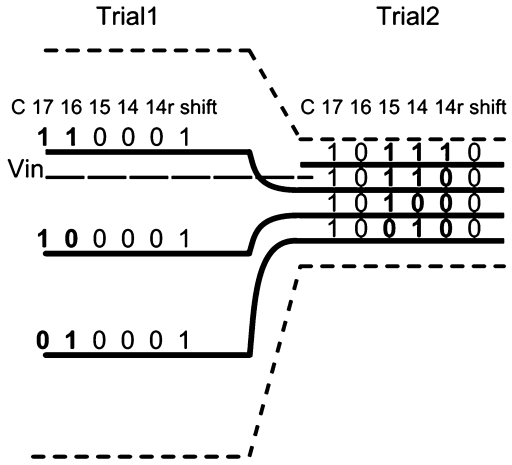


Fig. 8. Example ADC1 test levels for first 2 bit trials.

the negative shift provided by switching C_{shift} to V_{refn}, allow the input to be tested against the new test levels also shown in Fig. 8. The negative shift provided by C_{shift} is required to create a bipolar correction range.

For the third bit trial (not shown) C15 and C14 will be switched to V_{refp} and C14r to V_{refn}. After the 5th bit trial, when 10 bits will have been determined, the ADC1 smallest capacitors associated with the 5th bit trial are also set to the same state and the resulting identical residues from all 4 DACs are connected via the S1 switches to the input of the RA. Since all four sampling DACS are switched to the RA, the sampled thermal noise corresponds to the total capacitance of all four sampling DACs.

One of the key architectural decisions was how many bits to do in ADC1. Fewer bits allow the offset specification for the ADC1 comparators to be relaxed. More bits give a smaller ADC1 residue which allows a higher RA gain and therefore a larger ADC2 LSB size. A larger ADC2 LSB size then relaxes the offset and noise requirements for the ADC2 comparators. The decision to do 10 bits within ADC1 was predicted to give the lowest power solution.

B. Residue Amplifier

A simplified schematic of the RA is shown in Fig. 9. The RA has a high closed loop gain of 42.66, set by the ratio of the total capacitance of all the sampling capacitors in the 4 ADC1 DACs (256 unit capacitors) to that of the feedback capacitor C1 (6 unit capacitors). This gain was the largest achievable with unit sized feedback capacitors that did not saturate the RA output stage. The corresponding feedback factor is small and is approximately equal to the reciprocal of the gain. To help compensate for this low feedback factor, the open loop gain of the RA is increased by preceding the transconductance amplifier with a preamp set to a gain of 20. The resulting open loop bandwidth is sufficient to provide full settling of the RA within 18 ns. The stability of the amplifier is very sensitive to the gain of the preamp. Increasing the preamp gain reduces the frequency of the main

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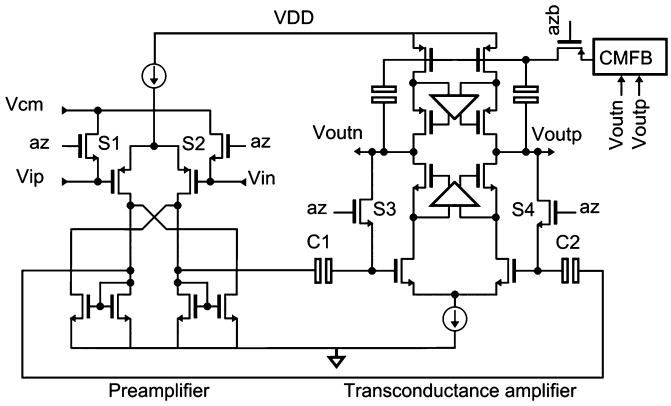


Fig. 9. Residue Amplifier simplified schematic.

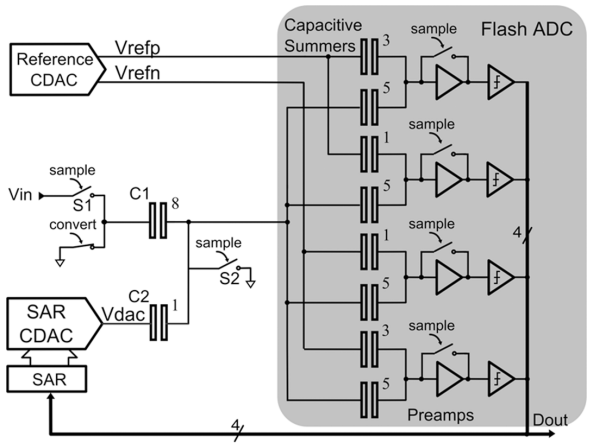


Fig. 10. ADC2 block diagram (actual implementation is differential).

non-dominant pole at the output of the first stage and at the same time increases the open loop bandwidth. Beyond a gain of 20, the reduction in stability starts to cause an increase in simulated settling time.

Auto-zeroing is used to reduce offset and 1/f noise. During an auto-zero phase, switches S1 and S2 connect the input to a common mode voltage and switches S3 and S4 short the outputs of the transconductance amplifier to its inputs. Opening these switches at the end of the auto-zero phase captures both the output referred offset of the preamp and the input referred offset of the transconductance amplifier on the inter-stage capacitors C1 and C2.

The RA nonlinearity and gain error are new error sources that do not exist in a traditional SAR ADC. However, since a full ten bits are determined within ADC1, the full scale range at the RA output is only 9 bits (allowing 1 bit of overlap across the RA). This considerably eases the gain and linearity requirements of this amplifier. The closed loop gain of the RA is factory trimmed. To ensure that this gain remains sufficiently stable with temperature, gain boosted cascodes have been used within the transconductance amplifier.

In a traditional SAR ADC, the dominant noise sources are the comparator preamp and the input sampling DAC kT/C sampling noise. In this architecture the input sampling DAC noise contribution $\sim 2/3$ LSB rms remains the same but the other dominant noise source is now the RA ~ 2 LSB rms: As with a more conventional pipeline architecture the comparators only need provide coarse results.

C. ADC2

Fig. 10 shows a simplified block diagram of ADC2 which converts the output of the RA. This architecture minimizes the loading on the RA by using a single capacitor C1 to sample the output of the RA. The loading on RA is further reduced by using a small value for C1. This can be tolerated since kT/C noise sampled on to this capacitor (and other ADC2 capacitors), when referred to the ADC input, is divided by the gain of the RA. The architecture is based on [11] and uses just 2 DACs: a SAR DAC whose output (scaled by -8) converges on $\mathrm{V_{in}}$ through the conversion; and a Reference DAC that provides the reference for a 4 level flash ADC. Both DACs use the same reference voltage as the DACs within ADC1. C1 along with associated switches S1–S2 is used to sample the input. During conversion the sampled value is capacitively combined with the output of the successive approximation DAC by capacitors C1 and C2 to produce a signal proportional to $\mathrm{V_{DAC}}/8 - \mathrm{V_{in}}$ at the input to the flash ADC. The ratio of 8:1 between C1 and C2 is required to match the LSB size of the successive approximation DAC to the LSB size at the output of the RA. The flash ADC result is used to set the Successive Approximation DAC output $\mathrm{V_{DAC}}$ to the new trial value for the next bit trial. In a traditional SAR ADC, the resolution of each subsequent trial increases by a factor of 2. Since 2 bits are being determined each trial, the resolution must increase by a factor of 4. To achieve this increase in resolution, the reference DAC output (after an initial full scale step for the first bit trial) decreases by a factor of 4 each subsequent trial. Four differently weighted capacitive summing networks are used to combine the outputs of the 2 DACs. The output of each summer is sensed by a comparator. At the flash ADC code transitions, the differential voltage at one of the 4 comparator inputs will be zero. The following relationships apply at these 4 code transitions:

$$
\begin{aligned}
&0 \text{ to } 1 \text{ transition: } (\mathrm{V_{DAC}}/8 - \mathrm{V_{in}}) * 5/14 = 3/8\ \mathrm{V_{ref}}\\
&1 \text{ to } 2 \text{ transition: } (\mathrm{V_{DAC}}/8 - \mathrm{V_{in}}) * 5/14 = 1/8\ \mathrm{V_{ref}}\\
&2 \text{ to } 3 \text{ transition: } (\mathrm{V_{DAC}}/8 - \mathrm{V_{in}}) * 5/14 = -1/8\ \mathrm{V_{ref}}\\
&3 \text{ to } 4 \text{ transition: } (\mathrm{V_{DAC}}/8 - \mathrm{V_{in}}) * 5/14 = -3/8\ V_{\mathrm{ref}}.
\end{aligned}
$$

The architecture therefore allows the 'error' voltage $(\mathrm{V_{DAC}}/8 - \mathrm{V_{in}})$, representing the difference between the new trial $\mathrm{V_{DAC}}$ value (scaled) and Vin, to be tested against 4 equally spaced test levels. While only 3 levels are required to determine 2 bits, the fourth level, as in ADC1, provides some redundancy.

D. Dither

While offsets in the ADC2 auto-zeroed comparators were predicted to be at about the sub LSB level, these offsets will still cause uneven code transitions and degraded DNL. To avoid this degradation in DNL, dithering has been used within ADC2. Dithering randomly shifts the ADC transfer function from one

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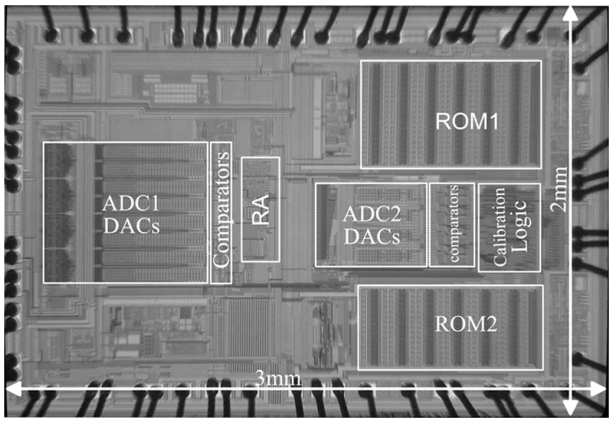


Fig. 11. Die photograph.

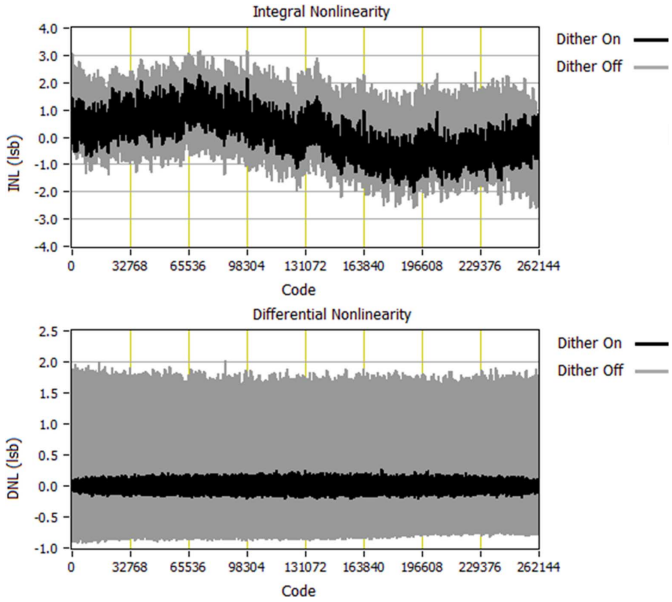


Fig. 12. Measured INL and DNL at 12.5 MS/s with dither turned both on and off.

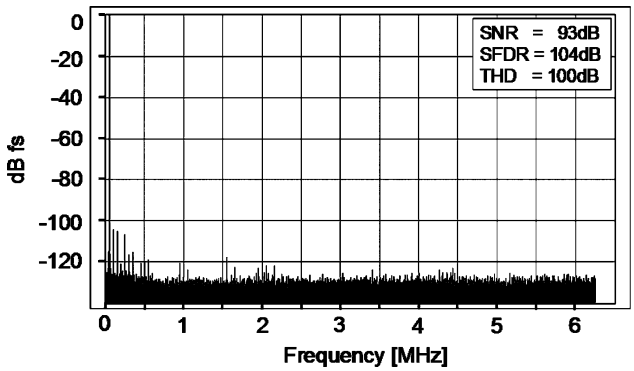


Fig. 13. Measured FFT with ADC sampling a 50 kHz input at 12.5 MS/s.

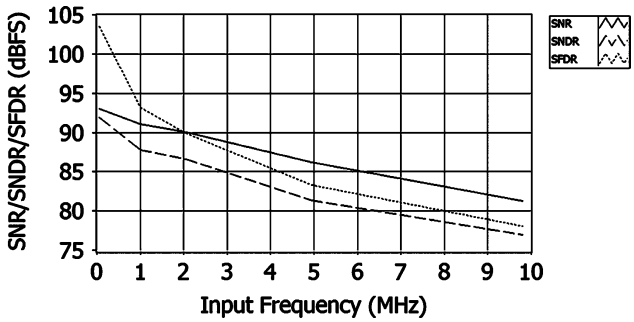


Fig. 14. Measured SNR, SNDR and SFDR at 12.5 MS/s vs input frequency.

conversion to the next so that the offset of a particular comparator will, over a number of conversions, affect all code transitions equally [12]. The dither value is generated by a 5-bit pseudorandom binary sequence generator and is applied to a sub-DAC of the SAR DAC within ADC2. To avoid the dither perturbation used to shift the ADC2 transfer function directly degrading SNR, the dither value added to the sub-DAC is subtracted from the digitally corrected result, prior to clipping to produce the final result. A gain error in this sub-DAC will result in a mismatch between the dither added in the analog domain and the dither subsequently subtracted in the digital domain. This mismatch creates an error known as dither leakage which degrades SNR. Since the dither full scale range is just 31 LSB the gain accuracy of the sub-DAC resulting from raw capacitor matching is sufficient to avoid any significant dither leakage and consequent reduction in SNR. The effect of dithering on DNL is described in the Measurement Results section.

V. Measurement Results

The 18-bit pipeline ADC, consisting of a pipeline of 2 multi-bit SAR ADCs, was implemented on a 0.25 μCMOS process using PIP capacitors and both 2.5 V and 5 V PMOS and NMOS devices on a single monolithic die. The 5 V devices were used mainly in the input circuits and for the DAC switches and associated drivers to withstand voltages between 0 V and 4.1 V in normal operation. Power consumption is 105 mW, excluding LVDS interface power. A photograph of the die is shown in Fig. 11. The die measures 3 mm by 2 mm. All reported measurements were made with the ADC signal inputs each driven by an external op-amp configured as a unity gain follower.

Histogram testing with a 15 Hz sinusoidal input was used to measure INL and DNL with an average of 250 hits per code. INL and DNL with and without dithering are shown in Fig. 12. Without dithering INL is $+3.1/-2.7$ LSB and DNL is $+2/-0.95$ LSB. With dithering INL is improved to $+2.2/-2.0$ and more significantly, DNL is improved to ± 0.3 LSB.

Fig. 13 shows an FFT for the ADC operating at 12.5 MS/s with a 50 kHz input tone. SNR is 93 dB SFDR is 104 dB. Fig. 14 shows SNR, SNDR and SFDR over a range of input signal frequencies. The ADC achieves an SNR of 93 dB at 50 kHz but this degrades to 91 dB at 1 MHz and 81 dB at 9.8 MHz.

A figure of merit (FOM) allowing a fair comparison of the power (P) efficiency with which bandwidth (BW) and dynamic

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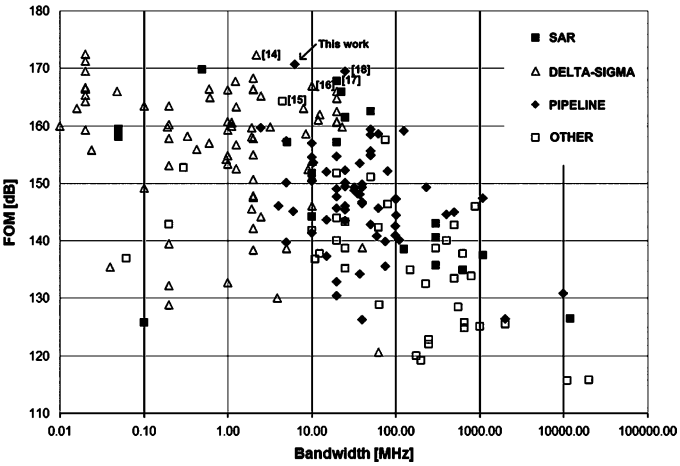


Fig. 15. FOM of ADCs presented at ISSCC from data in [14].

range (DR) are achieved is proposed and justified in [13]. The FOM is defined as

$$\mathrm{FOM} = \mathrm{DR} + 10\log\left(\frac{\mathrm{BW}}{P}\right).$$

For Nyquist rate converters, BW is half the sample rate. A scatter plot of FOM vs. bandwidth of ADCs presented at ISSCC from 1997 to 2010 is shown in Fig. 15, [14]. This ADC achieves the highest FOM for a converter with a BW greater than 2.2 MHz.

VI. Conclusion

An 18-bit 12.5 MS/s converter consisting of a pipeline of two multi-bit SAR ADCs has been presented. The ADC achieves a SNR of 93 dB with a 50 kHz input signal and INL and DNL better than ± 2.2 LSB and ± 0.3 LSB respectively. These performance metrics are similar to those of commercially available precision SAR ADCs operating at less than one third the output data rate of this converter. The good DC linearity achieved with this converter is believed to be due, at least in part, to the relatively small RA full-scale range of 9 bits. This small full-scale range avoids gain stability and linearity errors of the RA translating into significant ADC linearity errors. Dither has been shown to be highly effective in reducing DNL errors created by offsets in the flash ADC used within the second SAR ADC. SFDR degrades faster with input frequency than was predicted by simulation, but is comparable with many other precision SAR ADCs. The key performance metrics of SNR and INL, required for digital X-ray systems, are achieved with some margin at a conversion rate of 12.5 MS/s.

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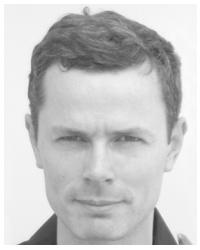
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